Which Leaderboard Orderings Survive a Human Reader?

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Abstract

A benchmark score is produced by a model and by a verifier that decides what answer the model's text contains. Only the model is named in the leaderboard. We ask what happens to a published ranking when a human reader, rather than a string matcher, decides what each response says.

On the Open LLM Leaderboard's MATH-Hard board, we read all 35,748 published responses from 27 models with three verifiers: the leaderboard's original 2024 scorer, the verifier behind its February 2025 re-scoring, and a boxed-answer extractor following the benchmark's own reference convention. Re-executing the scoring function recorded in the leaderboard's result files reproduces its published per-item verdicts on 35,748 of 35,748 responses, which establishes the identity of the published verifier from records rather than from agreement. The three verifiers disagree on 1,159 responses, and those responses are where the ranking is decided.

Certifying a ranking from a few hundred human labels requires pooling across models, and pooling requires an assumption about how far the models may differ where the verifiers agree. We state that assumption as a single parameter ε, derive an identified range for each pairwise difference under it, and report certified orderings as a function of ε instead of at one chosen value. Of 351 pairwise orderings, 153 are certified under exact homogeneity, 131 at the tolerance we registered before seeing results, and 0 with no pooling at all. The audit bounds are design-valid without assuming equal rates across models, and simulation at the full 27-model scale confirms the stated coverage where the assumption holds and shows it failing where it does not.

Two provenance findings shape what can be claimed. The leaderboard changed its own MATH verifier in February 2025, so the per-item verdicts it publishes and the scores it displays come from different scorers. And our own boxed verifier was silently degraded by a platform-specific defect that reduced symbolic equivalence to string comparison; correcting it moves the count of disagreeing responses on our earlier subset from 817 to 704. We report the corrected analysis together with the defect that produced the earlier numbers.

We release the human labels, the per-cell census of all 35,748 responses, the three verifier configurations with their shared components stated explicitly, and the code that regenerates every number in this paper.

1 Introduction

A model was asked how many more euros than pounds Kathy has. It converted both amounts, compared them, and wrote: "Kathy has $\boxed{22}$ more euros than pounds. Final Answer: The final answer is 22. Great job!" The reference answer is 22. A person reading the response would mark it correct. The leaderboard recorded it as wrong.

Nothing went wrong with the model. The scorer that produced that verdict looks for one sentence pattern, "Final Answer: The final answer is X. I hope it is correct.", and this response ended with "Great job!" instead. The extraction returned its sentinel for no answer found, and a response with no answer found is recorded

exactly as a response with a wrong answer. The published score for that model is one item lower, and nothing in the published output says why.

This is not a rare accident, and it is not confined to one scorer. Every benchmark score is produced by two components. A model writes text, and a *verifier* decides what answer that text contains and whether it matches the reference. Only the first component is named in the leaderboard. The second is a configuration choice that varies between frameworks, between tasks inside one framework, and over time within a single task: the Open LLM Leaderboard re-scored its own MATH results in February 2025 with a different verifier, and the published numbers changed while the model outputs stayed exactly as they were.

The consequence we care about is not that scores are slightly too low. Everyone who has looked at this already knows that a benchmark score is a lower bound on what a careful reader would credit, and a uniform downward bias would be a nuisance rather than a problem. The consequence is that the *ranking* can move. Lower bounds do not order models. If one model happens to phrase its answers the way the verifier expects and another does not, the gap between them is partly a fact about formatting conventions and partly a fact about mathematics, and the published number does not separate them.

What this paper does. We take the 27 models of the Open LLM Leaderboard's MATH-Hard board, with all 1,324 Level-5 items and the leaderboard's own published generations, and ask which of the 351 pairwise orderings survive when a human reader, rather than a string matcher, decides what each response says.

Three verifiers read the same responses. The first is the leaderboard's original 2024 scorer, whose per-item verdicts are published in its own result files. The second is the verifier behind the leaderboard's current published scores, introduced in the February 2025 re-scoring. The third extracts the last boxed expression, which is the convention the benchmark's own reference solutions use. Each response therefore falls into one of eight *cells*, according to which verifiers credited it. The cells where all three agree cover most of the board. The cells where they disagree, 1,159 of 35,748 responses, are where the ranking is decided.

Human labels are expensive, so they go where the disagreement is. We reuse an existing 400-item audit and add a preregistered 458-label audit stratified by model and cell. The estimate that the audit supports is not a correction to each model's score. It is a bound on each pairwise difference, and whether that bound excludes zero.

The assumption, stated in advance. A small audit cannot measure every model separately in every cell. Certifying a ranking from a few hundred labels requires pooling, and pooling requires an assumption: that in the cells where the verifiers agree, the rate at which a human reader overrules them does not vary too much across models. We write this as $H(\varepsilon)$, where ε bounds the spread across models, and we report results as a function of ε rather than at a single convenient value.

At $\varepsilon = 0$, which is exact homogeneity, the existing audit certifies 153 of 351 orderings. At the tolerance we registered before seeing any result, $\varepsilon = 0.01$, it certifies 131. Without pooling at all, it certifies 0. The assumption carries the result, and saying so plainly is part of the contribution rather than a caveat attached to it.

We also report what $\varepsilon = 0$ actually means. Because ε constrains realized fractions over finitely many responses, exact homogeneity is unattainable whenever the underlying rate is strictly between zero and one; sampling variation alone spreads the per-model fractions. In simulation at the full scale of this board, an analysis conducted at $\varepsilon = 0$ against a world with homogeneous underlying rates exceeded its nominal error rate, and a tolerance of 0.01 restored it. We therefore treat $\varepsilon = 0$ as an idealization that belongs at one end of a curve, and never as a headline.

What we found along the way. Two findings emerged from checking the provenance of the verdicts rather than from the analysis itself, and both change what can be claimed.

First, the leaderboard's published per-item verdicts are reproducible. Re-executing the scoring function recorded in the leaderboard's own result files reproduces the published verdict on 35,748 of 35,748 responses.

The identity of the published verifier is therefore established from the leaderboard's records rather than inferred from agreement rates.

Second, an earlier version of our own boxed verifier was silently degraded on Windows. The symbolic equivalence check it relies on guards itself with a Unix-only timer, and on Windows it failed and returned "not equivalent" for every call it was asked to make. Every verdict that verifier produced before we found this was a string comparison. Correcting it changes the number of disagreeing responses among the models we had analysed previously from 817 to 704, and the certified count from 84 to 89. We report the corrected analysis, and we report the defect, because a reader deciding whether to trust a verifier audit is entitled to know that the authors of one found a bug of exactly the kind the paper is about, in their own pipeline.

What we do not claim. We do not claim that any verifier here is correct in an absolute sense, and we do not treat the human labels as ground truth beyond what they are, which is a careful reader's judgement of what a response commits to. The three verifiers share a normalization routine and a symbolic algebra library, so their agreement is not statistical independence, and we give the shared components explicitly rather than describing the verifiers as independent. The audit screens a small number of models for substantial departures from the pooling assumption and cannot detect small ones at this budget. Finally, everything here is measured on one benchmark and one leaderboard; the harness-level evidence in Section 6 speaks to how widely the same design choice is made, not to how often it changes a published ranking elsewhere.

2 Setup and identification

2.1 What is fixed, what is observed, and what is unknown

The benchmark is fixed. It consists of 1,324 items, each with a reference answer, and for each of 27 models the leaderboard publishes one response per item. No model is run again in this paper, and no response is regenerated. Everything below is a statement about those published responses.

A *verifier* is a function that reads a response and the reference answer and returns either 1, meaning the response should be credited, or 0. A verifier has two parts that can each fail. It extracts a candidate answer from free text, and it decides whether that candidate is equivalent to the reference. The first part fails when a response states its answer in a form the extraction rule does not recognise, and a response whose answer cannot be extracted is scored 0, which is the same value a wrong answer receives.

We apply $K = 3$ verifiers to every response, so each response carries a vector of K verdicts and falls into one of 8 *cells*. We write a cell as the string of verdicts in a fixed order, so that cell 111 contains the responses every verifier credits and cell 000 contains the responses no verifier credits. We call those two cells *concordant* and the remaining cells *discordant*. Table 1 gives the sizes.

The *human reference* Y^* is defined for every response, although it is observed only for audited ones. A labeler reads the response, without seeing the reference answer or any verifier's verdict, and transcribes the final answer the response states, exactly as written. A labeler who finds no final answer records NONE, and a labeler who finds two different final answers both presented as final records CONTRADICTORY. We then set $Y^* = 1$ when that transcription matches the reference answer under the benchmark's own equivalence routine, and $Y^* = 0$ otherwise, counting NONE and CONTRADICTORY as 0. Section E reports what changes when they are dropped instead.

Two quantities follow. The first is the *cell share* s_{mc}, the fraction of model m's responses lying in cell c. Shares are computed from all 35,748 responses and are known exactly, with no sampling error, because every response is read by every verifier. The second is p_{mc}, the fraction of model m's responses in cell c that the human reference credits. These are the unknowns. Each model's accuracy under the human reference is

$$A_m \;=\; \sum_c s_{mc}\, p_{mc}, \tag{1}$$

and the quantity a ranking claim is about is the pairwise difference $\Delta_{jk} = A_j - A_k$.

2.2 Why the audit alone does not identify the ranking

Equation (1) is linear in the unknown p_{mc}, and the coefficients are known. If every p_{mc} were known, every ordering would follow. An audit estimates them from a sample, and a sample of a few hundred labels spread over 27 models and 8 cells leaves most (m, c) pairs with no labels at all.

For a cell with no labels, the only honest statement about p_{mc} is that it lies in $[0, 1]$. Substituting that into (1) for both models gives a range for Δ_{jk} whose width is the combined share of the unaudited cells. Whenever that width exceeds the observed gap between two models, the ordering is not identified, and no amount of statistical machinery applied to the audited cells changes it. This is why the analysis is organised around cells rather than around models: the discordant cells are small, so bounding them crudely is affordable, while the concordant cells are large, so they must be bounded tightly or nothing is identified.

2.3 The pooling assumption

Tight bounds on the concordant cells require pooling labels across models, and pooling is an assumption rather than a measurement. We state it as a single parameter.

Assumption 1 (Bounded heterogeneity, $H(\varepsilon)$)**.** For each concordant cell c, the values of p_{gc} across the audit sub-populations g that make up that cell lie in an interval of width at most ε.

A sub-population is a model, or a model crossed with a design stratum where the audit design was stratified more finely. Assumption 1 restricts how far models may differ from each other in the cells where the verifiers agree. It says nothing about the level. The rate at which a human reader overrules a set of agreeing verifiers may be high or low, and the audit estimates it; ε bounds only the spread.

Exact homogeneity, $\varepsilon = 0$, is stronger than it looks. Each p_{gc} is a fraction over finitely many responses, so if the underlying rate is strictly between zero and one, two sub-populations of finite size will differ by chance alone. $H(0)$ can hold exactly only when the rate is zero or one. We therefore report certified orderings as a function of ε, mark the tolerance $\varepsilon = 0.01$ that we registered before seeing any result, and treat $\varepsilon = 0$ as the upper end of a curve rather than as an estimate.

2.4 The identified range

Let $[L_c, U_c]$ be a confidence interval for the pooled correctness in concordant cell c, constructed in Section 3 so that it is valid without assuming the models share a common rate. For a discordant cell, let $[l_{mc}, u_{mc}]$ be a per-model interval, which is $[0, 1]$ when that cell has no labels for that model.

Proposition 1 (Identified range)**.** *Fix a pair (j, k) and a concordant cell c. Under Assumption 1 and on the event $p_c \in [L_c, U_c]$, the pair (p_{jc}, p_{kc}) lies in*

$$R_c(\varepsilon) \;=\; \big\{(x, y) \in [\ell_c, h_c]^2 \;:\; |x - y| \le \varepsilon\big\}, \qquad \ell_c = \max(0, L_c - \varepsilon), \quad h_c = \min(1, U_c + \varepsilon).$$

Consequently Δ_{jk} lies in an interval whose endpoints are obtained by minimising and maximising (1) separately in each cell, and for a concordant cell with shares $a = s_{jc}$ and $b = s_{kc}$ the minimum of $ax - by$ over $R_c(\varepsilon)$ equals $a\ell_c - b\min(h_c, \ell_c + \varepsilon)$ when $a \ge b$, and $a\max(\ell_c, h_c - \varepsilon) - bh_c$ otherwise.

At $\varepsilon = 0$ the region collapses to the diagonal, $p_{jc} = p_{kc}$, which is the pooled analysis; the range is then sharp. For $\varepsilon > 0$ the region is an outer bound, because the weights with which j and k enter the pooled estimate constrain the attainable configurations slightly more than $R_c(\varepsilon)$ does. Using the outer bound is conservative in the direction that matters: it can only widen the reported range, never narrow it. Appendix A proves the proposition and the closed form, and Appendix B checks both against brute-force enumeration.

2.5 Certification and breakdown

An ordering $j \succ k$ is *certified* at ε when the identified range for Δ_{jk}, widened by the sampling term of Section 3, lies entirely above zero. Certification is monotone in ε, since $R_c(\varepsilon)$ grows with ε, which makes the following well defined.

Definition 1 (Breakdown value)**.** ε^*_{jk} is the largest ε at which the ordering $j \succ k$ remains certified.

The breakdown value answers the question a skeptical reader should ask, which is not whether an ordering is certified but how much cross-model heterogeneity it would take to overturn it. Reporting ε^* per ordering makes the dependence on Assumption 1 a measured quantity rather than a caveat. Because the range in Proposition 1 is an outer bound for $\varepsilon > 0$, each ε^*_{jk} is a lower bound on the true breakdown value, so the reported robustness is understated rather than overstated.

2.6 Why more than one verifier is needed

Stratifying on a single verifier does not help, and the reason is worth stating because it explains the design.

Proposition 2 (A common error mechanism cannot reorder)**.** *Suppose one verifier is used, and suppose its predictive values are common to all models, so that* $\Pr(Y^* = 1 \mid v = 1) = \pi_1$ *and* $\Pr(Y^* = 1 \mid v = 0) = \pi_0$ *regardless of* m*. Then* $\Delta_{jk} = (\pi_1 - \pi_0)(a_j - a_k)$*, where* a_m *is the rate at which the verifier credits model* m*.*

Every pairwise difference is scaled by one common factor. A common error mechanism therefore cannot produce a reordering of one pair while leaving another intact; its only global effect is to reverse every ordering at once, when $\pi_1 < \pi_0$. Pair-specific reversals, which we observe, require the mechanism to differ across models. Pooling under a single verifier assumes away exactly the phenomenon under study, whereas pooling restricted to the cells where independent verifiers agree leaves the disagreement cells free to be model-specific.

3 Inference

3.1 An interval that does not assume the models are alike

The audit draws labels by stratified sampling without replacement, and the pooled count in a concordant cell sums labels from sub-populations whose true rates may differ. A textbook binomial interval is exact for a single rate under sampling with replacement, and neither condition holds here. Using one anyway would assume, inside the confidence statement, the very homogeneity that Assumption 1 is supposed to make explicit and testable.

We use a bound that survives both departures.

Lemma 1 (Design-valid pooled bound)**.** *Let* x_c *be the total number of audited responses credited by the human reference in cell* c*, summed over sub-populations, with* n_c *labels in total and sampling weights* $w_{gc} = n_{gc}/n_c$*. Let* $\tilde{p}_c = \sum_g w_{gc}\, p_{gc}$*. Then for every* $t > 0$*,*

$$\Pr\left(\frac{x_c}{n_c} \geq \tilde{p}_c + t\right) \leq e^{-n_c \mathrm{KL}(\tilde{p}_c + t \,\|\, \tilde{p}_c)}, \qquad \Pr\left(\frac{x_c}{n_c} \leq \tilde{p}_c - t\right) \leq e^{-n_c \mathrm{KL}(\tilde{p}_c - t \,\|\, \tilde{p}_c)}.$$

The proof, in Appendix A, combines two standard facts. Sampling without replacement is dominated by sampling with replacement in the sense required for moment generating functions, which removes the first difficulty, and the moment generating function of a sum of independent Bernoulli variables with different means is bounded by that of a single Bernoulli at the weighted mean, which removes the second. Inverting the bound gives an interval $[L_c, U_c]$ with coverage at least the nominal level, valid in finite samples, for heterogeneous rates, and without replacement.

Clopper–Pearson intervals are tighter, and we report them alongside as a sensitivity. They are exact only under assumptions this design does not satisfy, so the headline numbers use Lemma 1.

3.2 Item sampling

The benchmark's 1,324 items are themselves a sample of the questions one might have asked. A claim about which model is better in general, rather than on these items, carries a second source of uncertainty that the audit cannot reduce. We bound it by a paired argument. For a pair of models, let D_i be the difference of their human-reference verdicts on item i. Then $\mathrm{Var}(D) \leq \Pr(D \neq 0)$, and $\Pr(D \neq 0)$ is bounded above by

the observed rate at which a verifier's verdicts differ between the two models, plus each model's maximum possible disagreement with that verifier, both of which the identified range already supplies. The resulting half-width is paired, so it exploits the fact that both models answer the same items, and it is conservative, because it uses upper bounds rather than estimates.

This term is asymptotic. It rests on a central limit approximation, and we say so rather than presenting the combined interval as exact. Appendix C reports a distribution-free alternative, which is valid in finite samples and noticeably wider.

3.3 What is certified, and at what level

Two sources of error are combined by a union bound, and the two are distinct. The audit interval concerns the unknown correctness rates given these responses; the item-sampling term concerns generalisation beyond these items. Nothing is counted twice.

Families of claims are controlled explicitly. A statement about one ordering is made at the per-ordering level. A statement about the whole board is a statement about 351 orderings at once, so the sampling term is widened by a Bonferroni factor over the family, and the pooled concordant intervals are shared by every pair, which makes the audit part simultaneous automatically. Rank intervals follow: if every certified ordering is correct, model m's rank is at least one plus the number of models certified above it and at most the total less the number certified below it. We report rank intervals as the family-level object, because a reader looking at a leaderboard is reading all the orderings at once, not one.

Proposition 3 (Coverage)**.** *Under Assumption 1, all audit-only claims hold simultaneously with probability at least* $1 - \alpha_{\text{audit}}$*, exactly in finite samples. Claims that also generalise beyond the benchmark's items hold with probability at least* $1 - \alpha_{\text{audit}} - \alpha_{\text{sample}}$*, asymptotically in the number of items.*

The two halves of that statement are deliberately different in kind, and we do not smooth over the difference. The first is a finite-sample guarantee that follows from Lemma 1. The second inherits the central limit approximation of the sampling term.

3.4 Simulation

Simulation cannot establish a coverage theorem, and we do not use it as one. It checks whether the guarantee is attained, and how it degrades when the assumption fails. We simulate at the full scale of the board, 27 models and 351 orderings, with 2,000 replicates per scenario, drawing audits under the real design and the real cell structure.

Where Assumption 1 holds, no certified claim was wrong in any scenario, so the procedure attains its stated level and is conservative rather than tight. Where the truth carries more heterogeneity than the analysis allows for, the family-wise error rises to 0.0675, against a nominal 0 observed under a correct specification. A world in which every model shares the same underlying rate, so that the per-model fractions differ only by chance, pushed the family-wise error to 0.185 when analysed at $\varepsilon = 0$, and a tolerance of 0.01 restored it. That last scenario is the quantitative reason the paper does not report $\varepsilon = 0$ as a result.

4 Data, verifiers, and audit design

4.1 The board

We use the Open LLM Leaderboard's (Beeching et al., 2023; Fourrier et al., 2024) MATH-Hard (Hendrycks et al., 2021) results: 27 models, 1,324 Level-5 items across seven subjects, and the generations the leaderboard itself published, read from its per-model result repositories. Every model has a response for every item, so the analysis uses 35,748 responses with no missing data and no model dropped. A prespecified rule covered missingness before the data were assembled, and it did not fire.

4.2 Three verifiers, and what they share

The first verifier is the leaderboard's original scorer. Its per-item verdicts are published in the result files, and we did not have to guess its identity: the result files record the scoring function that ran, and re-executing that function reproduces the published verdict on 35,748 of 35,748 responses. It extracts an answer by matching one sentence pattern and compares symbolically.

The second is Math-Verify, the verifier behind the re-scoring the leaderboard applied to its MATH results in February 2025 (Kydlíček et al., 2025). The re-scoring rewrote the published scores while leaving the per-item verdict files untouched, so the leaderboard currently displays scores from one verifier and publishes item-level verdicts from another. Running Math-Verify ourselves reproduces the re-scored published aggregates in 168 of 175 model-subject cells, with the remainder differing by a single item.

The third extracts the last boxed expression, which is how the benchmark's own reference solutions mark their answers, and compares it symbolically after the same normalisation the harness uses.

These verifiers are not independent, and we do not describe them as such. Table 2 lists what they share: a normalisation routine, a symbolic algebra library, and the same reference answers. What differs is the extraction rule, the parser, and the source from which the reference answer is taken. Independence is not required for any result in this paper. What is required is that they fail on different responses, which is an observable property, and they do.

4.3 A defect in our own verifier

While checking provenance we found that the boxed verifier had been silently degraded on this machine. Its symbolic equivalence check guards itself with a timer available only on Unix. On Windows the guard raises, the exception is swallowed by a broad handler, and the function returns "not equivalent" for every call. Every verdict it had produced was therefore a string comparison.

The corrected verifier changes the count of responses on which it disagrees with the leaderboard's scorer, over the subset of models analysed in earlier work, from 817 to 704, and the certified count from 84 to 89. All results in this paper use the corrected verifier. We report the defect because it is the paper's own argument turned on its author: a verifier that fails silently produces a plausible number, and nothing in the output says otherwise.

4.4 Human labels

Labels come from two audits of the same population. The first is an existing 400-item audit whose design was registered before labelling; of its labels, 144 fall in the cell where every verifier credits the response and 223 where none does. The second is a 458 label audit designed for this paper and registered before the sheet was drawn.

Labelling is done in a purpose-built offline tool that shows one response at a time, blocks pasting, records the time spent on each item, and writes an export whose hash is recorded. Every batch passes an automated quality gate before it is used. The gate checks structure, timing, and the rate at which a typed answer exactly matches what a regular expression would have extracted, compared against the rate observed in the author's earlier audit. Batches that fail are excluded and retained for the record, and this has happened.

4.5 Where the labels go

The design question is not how many labels to buy but where to spend them, and the two purposes compete. Labels in the discordant cells narrow the identified range and certify more orderings. Labels in the concordant cells test Assumption 1, on which every certified ordering depends, but do not themselves certify anything.

We computed the trade-off before drawing the sheet, over a grid of allocations, and it is reported in Figure 4 as a projection rather than as a result. Two findings came out of it. Certified orderings are nearly flat across allocations, because the binding constraints are the pooled concordant bound and item sampling rather than

Table 1: Verifier cells on MATH-Hard. A cell is the string of verdicts of the three verifiers, in the order (leaderboard 2024, boxed, Math-Verify). The last column is the share of audited responses in that cell that the human reference credits.

cell	reading	responses	human credits
000	none credits	32,685	0/223
111	all three credit	1,904	142/144
101	verifiers disagree	448	–
011	verifiers disagree	349	16/33
001	verifiers disagree	273	–
100	verifiers disagree	40	–
110	verifiers disagree	34	–
010	verifiers disagree	15	–

Table 2: What the three verifiers share. They are not independent; agreement between them is informative only because their extraction rules and parsers differ.

component	leaderboard 2024	boxed	Math-Verify
answer extraction	sentence pattern	last boxed expression	LaTeX and expression parse
normalisation	harness routine	harness routine	latex2sympy2
equivalence	sympy	string, then sympy	Math-Verify
reference answer	dataset field	dataset field	parsed from the solution

discordant coverage; the whole 458 label budget moves the projected count from 131 to 138. And a screen powerful enough to detect a moderate departure from Assumption 1 consumes most of the budget for a small number of models.

Since certification is nearly insensitive to the split, the budget goes where it buys the most information about the assumption. The registered design screens the 2 models appearing most often in the orderings with the smallest breakdown values, with 39 labels in the cell where no verifier credits and 100 where all three do, sized for 80% power against a per-model departure of 0.2. The remaining 180 labels go to discordant cells in proportion to their size.

What this screen cannot do is stated with it. It does not detect departures smaller than 0.2 at that power, and it says nothing at all about the models it does not screen. The preregistered rejection rule compares each screened model's error lower bound against the upper bound of the pooled rate computed without that model's own labels, so that a model cannot inflate the bound it is tested against.

5 Results

5.1 Where the verifiers disagree

Table 1 gives the cell structure. The two concordant cells hold most of the board: 1,904 responses are credited by every verifier and 32,685 by none. The verifiers disagree on 1,159 of 35,748 responses, which is a small fraction of the benchmark and, as Section 2 explains, the part that decides the ranking.

Figure 3 shows what a human reader makes of each cell. Where all three verifiers credit a response, the human reference agrees in 142 of 144 audited responses. Where none of them credits it, the human reference credits 0 of 223. The concordant cells are therefore close to their verifiers, which is what makes pooling worth doing and also what makes the remaining spread hard to measure. In the discordant cells the human reference splits, and five of them carry no labels at all in the existing audit, which is what the registered audit in Section 4 is for.

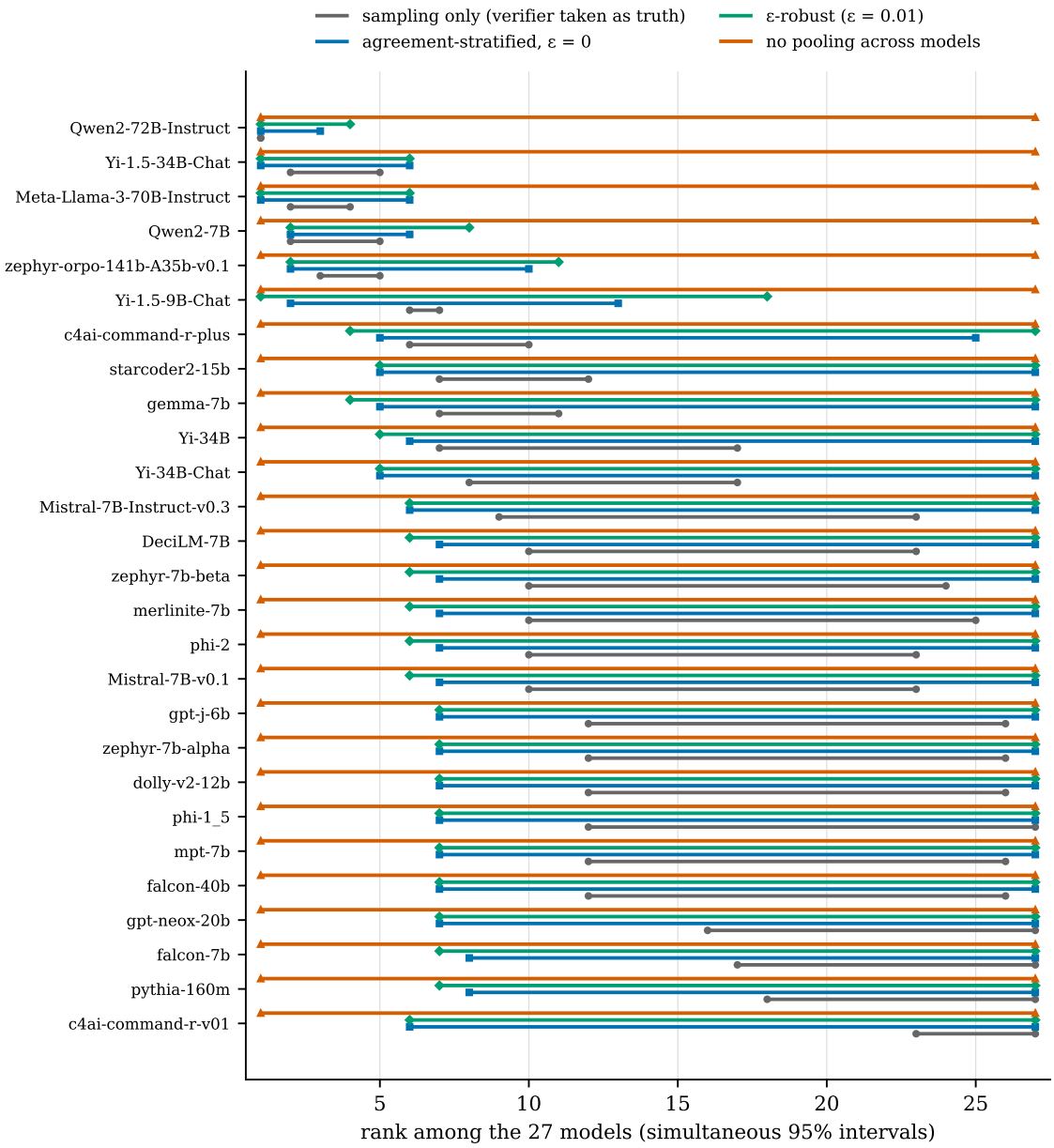


Figure 1: Simultaneous rank intervals for every model under four analyses. Taking the leaderboard's verifier as truth gives the narrowest intervals and is what a published leaderboard shows. Removing the pooling assumption entirely leaves almost every rank unresolved.

5.2 How many orderings survive

Of 351 pairwise orderings, item sampling alone would permit 280 to be called if the leaderboard's own verifier were taken as ground truth. That number is the ceiling against which everything else should be read, because no audit can certify an ordering that the benchmark is too small to resolve.

Under the existing audit, 153 orderings are certified at $\varepsilon = 0$ and 131 at the registered tolerance $\varepsilon = 0.01$. Requiring every claim on the board to hold simultaneously, rather than one ordering at a time, gives 109. With no pooling across models, which is the analysis that assumes nothing about how models relate to each other, 0 orderings are certified.

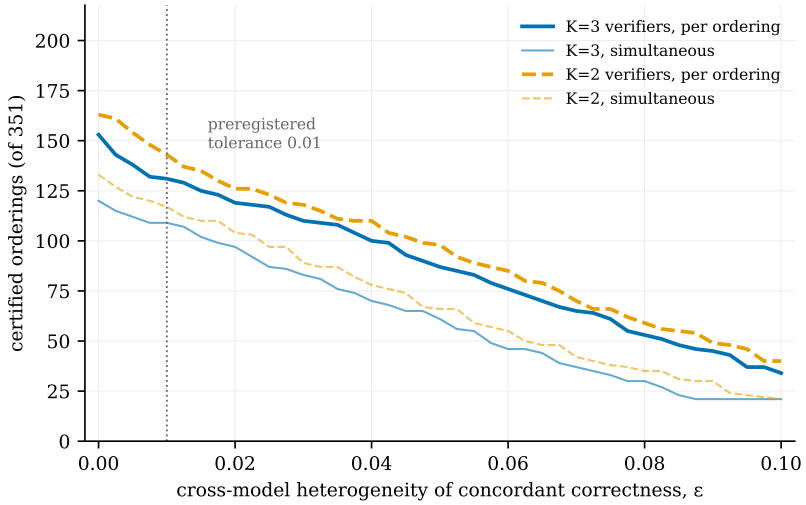


Figure 2: Certified orderings as a function of the permitted cross-model heterogeneity ε, for two and three verifiers, per ordering and simultaneously over the whole board. The dotted line marks the tolerance registered before results were seen.

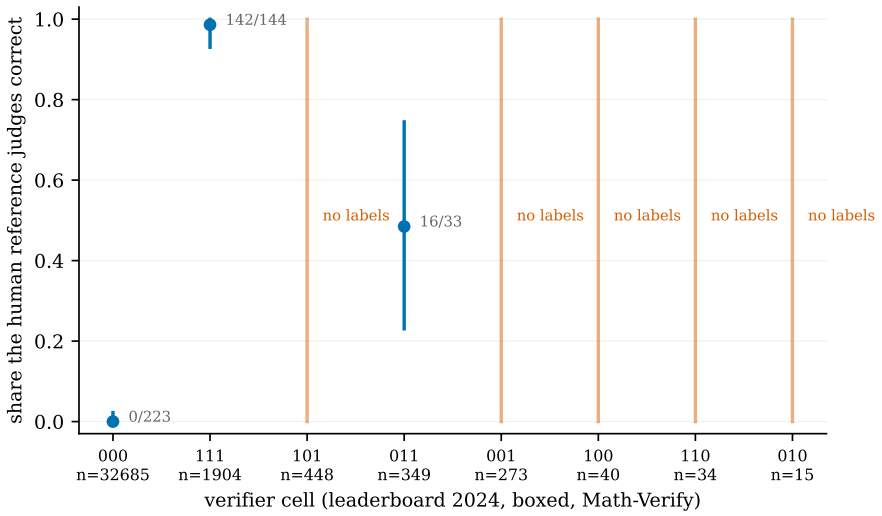


Figure 3: Share of audited responses in each verifier cell that the human reference credits, with intervals from Lemma 1. Five cells carry no labels in the existing audit and are bounded only by $[0, 1]$.

That last number is the honest centre of the paper. Nothing on this board is certified without an assumption connecting models to one another; the assumption is doing the work, and the only question worth arguing about is how much of it is required.

Figure 2 answers that question as a curve rather than a number. Certification declines smoothly as the permitted heterogeneity grows, and the median ordering breaks at $\varepsilon^* = 0.05769$. Half the certified orderings therefore survive a spread in concordant correctness several times larger than the tolerance we registered, and half do not.

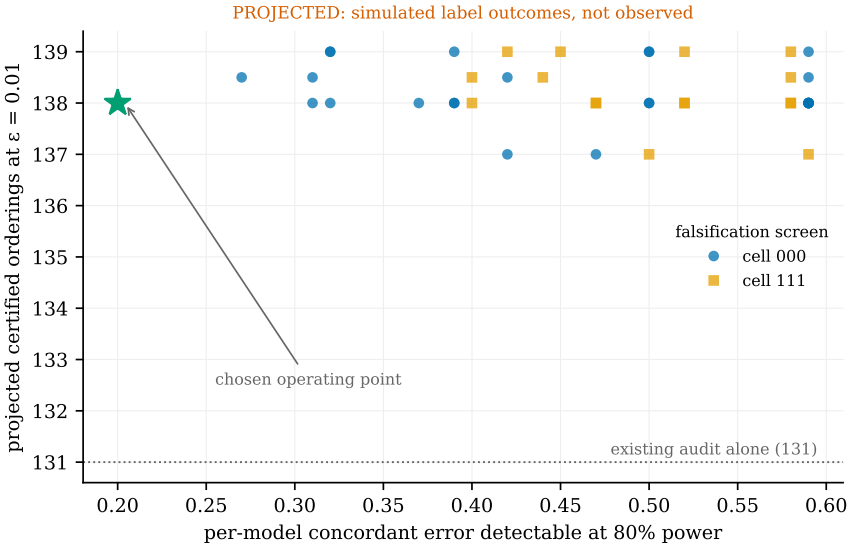


Figure 4: Projected trade-off between certifying orderings and screening the pooling assumption, over a grid of label allocations. Certification is nearly flat, so the budget is spent on the screen. Simulated label outcomes, not observed.

5.3 Ranks rather than pairs

Figure 1 reports simultaneous rank intervals under four analyses. Taking the leaderboard's verifier as truth gives the tightest intervals and is the status quo. The agreement-stratified analysis at $\varepsilon = 0$ widens them, the analysis at $\varepsilon = 0.01$ widens them further, and the analysis without pooling leaves nearly every model's rank unresolved across the whole board.

The comparison makes the cost of the published presentation visible. A leaderboard prints a rank without an interval, which is equivalent to the first analysis, and the first analysis is the one that assumes the verifier is never wrong.

5.4 Two verifiers or three

Using only the leaderboard's two scorers, rather than all three, certifies 163 orderings at $\varepsilon = 0$ and 143 at the registered tolerance, against 153 and 131 with three. Adding a verifier moves responses out of the concordant cells and into the discordant ones, where they are bounded more crudely, so a third opinion buys a weaker assumption at the price of a wider bound. Neither configuration dominates, and reporting both is the point.

5.5 Sensitivity

The conclusions above are unchanged under the alternatives we registered. Scoring indeterminate labels as incorrect rather than dropping them, using Clopper–Pearson intervals instead of Lemma 1, and substituting a Math-Verify-based human reference for the harness routine each move the certified count by a few orderings and change no qualitative claim. Appendix D gives the table.

The one substitution that matters is the defective verifier described in Section 4. Analysed with the string-only comparison it produced, the same audit certifies 84 orderings against 89 with the corrected verifier, on the subset where both can be computed.

5.6 The registered audit

[PENDING: the 458-label audit registered in Section 4 has been drawn but not yet labelled. On completion this subsection reports the falsification test for the 2 screened models, the certified counts with the new labels included, and the observed-versus-projected comparison in Figure 4. The projection is 138 certified orderings at $\varepsilon = 0.01$. No number in this subsection may be written before the labels exist.]

6 How common is this design choice?

One board is one board. The results above show that the verifier changes the ranking on MATH-Hard, and a reader is entitled to ask whether that is a peculiarity of this benchmark or of the frameworks that score it. The question splits into two, and only one of them is answered here.

The answerable question is how often a scoring configuration coerces an unreadable response into a wrong answer, which is the design choice that creates the problem in the first place. We resolved the task configurations of four evaluation frameworks (Biderman et al., 2024; Cao et al., 2026; Habib et al., 2023; OpenAI, 2024) with each framework's own loader at a pinned commit, and classified the scoring path each generative task actually takes. In `lm-evaluation-harness`, 2,936 of 4,524 generative tasks, or 64.9%, recover an answer by matching an assumed output format and record a failure to match as an incorrect answer. In OpenCompass the figure is 158 of 1,070, in `lighteval` 13 of 37 generative metrics, and in OpenAI's `simple-evals` 4 of 6 string-matching modules.

Reading code is not the same as running it, so the classification was validated behaviourally. For a random sample of tasks the declared filter pipeline was rebuilt from the framework's own classes and run on a well-formed response and on an equivalent response carrying the same answer in ordinary prose. Of 93 testable sampled tasks, all 93 behaved as classified, and of 100 control tasks drawn from the tasks classified as unaffected, all 100 were confirmed unaffected.

The unanswerable question, with the evidence in this paper, is how often that design choice changes a published ranking on some other benchmark. It depends on how often real responses take a form the extractor misses, which varies by benchmark, by model, and by prompt, and we measured it only here. The prevalence figures above bound how many tasks could be affected. They do not establish that any particular one is, and we do not claim otherwise.

The strongest statement the evidence supports is narrower than the one a reader might expect, and worth stating in full. On one board, with the leaderboard's own published generations, the choice of verifier moves the ranking. The design choice that makes this possible is the default across the major open evaluation frameworks. Whether it bites elsewhere is a measurement someone has to make, and the census says where to look first.

7 Related work

Whether a benchmark measures what it claims. One line of work treats benchmarks as instruments to be assessed rather than scoreboards to be topped. BetterBench grades benchmarks against criteria for design, implementation and documentation, and finds that many widely used ones score poorly (Reuel et al., 2024). A complementary framework asks which evaluative claims a given piece of evidence actually licenses, distinguishing performance on a narrow test from the broad capability the performance is often taken to demonstrate (Salaudeen et al., 2025). That work sits upstream of this one. It asks whether the questions and the construct are right; we take the benchmark and its reference answers as given and ask whether the ranking the pipeline publishes is identified by the pipeline's own outputs. The concerns compound rather than compete. A benchmark can be a valid instrument and still publish an unidentified ranking, and a benchmark whose construct is shaky can still be scored unambiguously.

Configuration moves scores, and extraction is part of the configuration. That evaluation results move with choices nobody reports is documented for prompt formatting (Sclar et al., 2023), for benign

perturbations that reorder leaderboards (Alzahrani et al., 2024), and across the whole surface of a widely used evaluation harness (Biderman et al., 2024). Answer extraction belongs to that surface. The pattern the leaderboard's original scorer matches descends from the convention introduced with Minerva (Lewkowycz et al., 2022), and proposals for better extractors, including learned ones, follow (Yu et al., 2025). The board studied here supplies its own demonstration: its MATH results were rescored with a different verifier in February 2025, and the published scores changed while the per-item verdict files did not (Kydlíček et al., 2025). Our claim is not that a better extractor exists. It is that no extractor is neutral, that the choice is currently unreported alongside the score, and that its consequence for a ranking can be bounded rather than argued about. A better extractor changes the cell structure; it does not remove the question of what happens where extractors disagree.

Verifiers as objects of measurement. Recent work measures verifiers directly, building benchmarks for verifier accuracy across domains (Li et al., 2025) and training verifiers meant to be robust to the formats that defeat regular expressions (Liu et al., 2025). That work and this one point in opposite directions from the same observation. Given labels, one can rank verifiers and adopt the best; we ask what survives when no verifier is adopted as truth, which is the situation a reader of a published leaderboard is in. The two are complementary: a better verifier shrinks the discordant cells, and the analysis here says what that shrinkage buys in certified orderings.

Partial identification when the label is corrupted. The bounds in Section 2 are partial-identification statements in the sense of Manski (2003). The closest classical results concern exactly this structure: what can be learned about a distribution when an unknown fraction of the data is contaminated (Horowitz & Manski, 1995), and the sharp bounds available when a binary variable is misclassified at unknown rates (Molinari, 2008). A verifier is a misclassifying measurement of correctness whose error rate is unknown and may depend on the model, so the same machinery applies with the models playing the role of sub-populations.

Concurrent work applies partial identification directly to language-model labels. Chen et al. (2026) study the prevalence of a latent binary attribute when the only evidence is a panel of model reports whose errors may be arbitrarily dependent given the truth, show that the prevalence is completely unidentified without restrictions that separate the latent components, and obtain bounds under weak stochastic-ordering restrictions. The diagnosis is the same as ours, and the resolutions differ in a way worth stating. They use replication of the reports and a stochastic-ordering restriction, and target a single prevalence with no ground truth at all. We use a human reference on a small audited subsample and a bounded-heterogeneity restriction across models, and target differences of prevalences, because a leaderboard reader consumes orderings rather than levels.

Reporting how far an assumption can be pushed. Assumption 1 is not testable at this budget in any strong sense, so the useful thing to report is how much of it a conclusion needs. That is the move made by design sensitivity in observational studies, where a finding is described by the magnitude of unmeasured bias that would be required to explain it away, and designs are compared by how much bias they can tolerate (Rosenbaum, 2004). The breakdown value of Definition 1 is the same object in this setting, with cross-model heterogeneity in place of unmeasured confounding: ε^*_{jk} is the amount of heterogeneity it would take to overturn the ordering $j \succ k$. The analogy also sets the reporting standard we follow. A sensitivity parameter is worth little as a caveat in a discussion section and worth a great deal as a number attached to each conclusion.

Two-phase sampling, verification bias, and raters without a gold standard. Auditing a cheap classifier with a small expensive sample is old. Two-phase sampling, in which a large first-phase sample is measured cheaply and a small second phase is measured accurately, goes back to Neyman (1938), and the binomial case with misclassification, where a fallible classifier is calibrated against a subsample scored by an infallible one, is Tenenbein's double sampling scheme (Tenenbein, 1970). The complication that the subsample is chosen by the fallible classifier's own verdict is verification bias, and corrections for it are standard in diagnostic testing (Begg & Greenes, 1983); confidence limits that propagate uncertainty in the estimated error rates into the corrected prevalence are also available (Lang & Reiczigel, 2014).

Two features of our setting break the classical correction. The human reference is applied to a sample stratified by the agreement pattern of several verifiers rather than by one verifier's verdict, and the error rates are not assumed common across the populations being compared, which is precisely the assumption Proposition 2 shows to be fatal to the question. The multi-rater literature is closer on the first point. Estimating error rates from several imperfect tests without a gold standard is classical (Hui & Walter, 1980), as is estimating observer error rates from repeated labels (Dawid & Skene, 1979). Those methods identify rates by assuming the raters err independently given the truth. We do not assume that, and Section 4 reports what the three verifiers share; instead of buying identification with an independence assumption, we buy partial identification with agreement stratification and pay for the rest in interval width.

Label-efficient evaluation with a proxy. Prediction-powered inference combines an abundant machine-generated proxy with a small human-labeled sample to produce valid confidence intervals for a target parameter, using the labeled sample to correct the proxy's bias (Angelopoulos et al., 2023). The family has grown towards evaluation: stratified variants for hybrid evaluation of language models (Fisch et al., 2024), active designs that spend the labeling budget where the proxy is least certain (Zrnic & Candès, 2024), and corrections for evaluation against synthetic labels (Boyeau et al., 2024). Two papers target our object directly. Chatzi et al. (2024) rank models from pairwise comparisons produced by a strong model, calibrated against a small set of human comparisons, and return rank sets with a coverage guarantee. Lee et al. (2025) correct the bias a judge's imperfect sensitivity and specificity induce in a naive score, build intervals that account for both the test set and the human-labeled calibration set, and allocate calibration samples adaptively.

The structural relationship is close and the difference is the assumption, not the arithmetic. Bias correction of this kind requires the proxy's operating characteristics, estimated on the calibration sample, to transport to the population being scored. When those characteristics are taken to be common across the models under comparison, which is the usual calibration setup, Proposition 2 applies: every pairwise gap is scaled by one number, and a pair-specific reversal cannot occur. The reversals reported in Section 5 therefore sit outside that model. Our response is to stop correcting and start bounding. Homogeneity is confined to the cells where independent verifiers agree, it is carried as a parameter that is varied rather than a condition that is assumed, and the output is a range with a breakdown value instead of a point estimate with an interval. The price is honest and large: far fewer conclusions, each of them conditional on a number the reader can disagree with.

Where to spend a label budget. The allocation question in Section 4 is a stratified sampling problem, and the classical answer optimises the variance of an estimated mean (Neyman, 1934), with unequal selection probabilities handled by the standard weighting (Horvitz & Thompson, 1952) and the finite-population corrections used here taken from the standard texts (Cochran, 1977; Kish, 1965). Adaptive versions that learn the within-stratum variances while sampling are current (Morikawa et al., 2026). Our allocation is not a variance problem. The two uses of a label, narrowing an identified range and testing the assumption on which every certified ordering depends, are not commensurable, and the projection in Figure 4 shows that the first is nearly flat in the allocation. The budget went where variance minimisation would not have sent it.

What this paper adds. No estimator here is new, and the individual ingredients are all borrowed from the work above. The contribution is the combination and the discipline: verifier provenance established from the leaderboard's own records rather than inferred from agreement, agreement stratification across K verifiers so that pooling is confined to the cells where they agree, an identification result that turns the pooling assumption into a reported parameter with a breakdown value attached to every ordering, a design registered in advance that splits a small label budget between certifying orderings and testing that assumption, and a released audit with its quality gate and its failures.

8 Limitations

The pooling assumption is not testable at this budget in any strong sense. The registered screen covers 2 of 27 models and detects a per-model departure of 0.2 with 80% power. Smaller departures, and any departure

in the models not screened, remain outside its reach. This is why the results are reported as a curve over ε and why the no-pooling analysis is reported next to them.

The human reference is a careful reader's judgement of what a response commits to, not mathematical truth. A labeler who transcribes an answer faithfully has not checked whether the answer is right; correctness is then decided by the benchmark's own equivalence routine against its own reference answer. Errors in those reference answers propagate to every result here and to the leaderboard equally.

The three verifiers share a normalisation routine and a symbolic algebra library, so a response that defeats that shared machinery can be miscounted by all three at once and land in a concordant cell. Such a response would be invisible to the agreement stratification and would be caught only by the human labels in the concordant cells, which is one reason the registered design spends labels there.

The item-sampling term is asymptotic, and the coverage statement says so. A distribution-free version is available and is substantially wider.

Everything is measured on one benchmark, one leaderboard, and a fixed set of published generations. The census in Section 6 shows how widely the same scoring design is used; it does not show how often that design changes a ranking elsewhere.

Finally, the audit labels were produced by a small number of labelers, one of whom is the author. The tool records timing and blocks pasting, batches pass an automated gate before use, and a subsample is independently re-labelled, but the labels are not crowd-sourced and their reliability is measured rather than assumed.

9 Conclusion

A leaderboard reports one number per model and names only half of what produced it. The other half is a verifier, and on the board studied here the choice of verifier moves the ranking, which a published score gives a reader no way to detect.

What a small human audit can do about that is limited and worth stating precisely. It can certify part of the ranking, but only under an assumption about how models relate to one another where the verifiers agree; without that assumption, this audit certifies 0 of 351 orderings. With it, at the tolerance registered before seeing any result, it certifies 131. The useful output is neither number on its own but the curve between them, together with the breakdown value of each ordering.

The practical recommendation is narrow and cheap. A benchmark that reports how often its scorer failed to read a response, alongside the score, would let a reader compute the bound in Section 2 without any audit at all. The count already exists inside the scoring pipeline; it is discarded.

Reproducibility statement

Every number in this paper is produced by a committed script from a committed result file and enters the text as a generated macro; a checker fails the build if any number is typed directly or any macro is undefined. The verifier configurations, the per-response census, the human labels, the preregistrations and their amendments, the quality-gate reports, and the code that regenerates every figure and table are released with the paper.

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A Proofs

Throughout this appendix a concordant cell c is partitioned into sub-populations indexed by g, where a sub-population is a model, or a model crossed with a design stratum when the audit design was stratified more finely inside the cell. Sub-population g contains N_{gc} responses, of which a fraction p_{gc} is credited by the human reference. The audit draws n_{gc} of them by simple random sampling without replacement, independently across g, and observes x_{gc} credited responses among those drawn. We write $n_c = \sum_g n_{gc}$, $x_c = \sum_g x_{gc}$, $w_{gc} = n_{gc}/n_c$, $\hat{p}_c = x_c/n_c$ and $\tilde{p}_c = \sum_g w_{gc}\, p_{gc}$, and $\mathrm{KL}(a \,\|\, b) = a \log(a/b) + (1-a)\log\big((1-a)/(1-b)\big)$ for $a, b \in [0,1]$.

Two things are worth saying before the proofs. The estimand $\tilde{p}_c$ is the *design-weighted* mean of the sub-population rates, not the population mean of cell c. Those coincide when the audit samples each sub-population in proportion to its size, and they do not in general. And $\tilde{p}_c$ is a quantity the design defines; the certification result uses it only through the interval $[L_c, U_c]$, together with Assumption 1, which ties every p_{gc} to every other.

A.1 Proof of Lemma 1

Proof of Lemma 1. Each x_{gc} is the number of credited responses in a simple random sample without replacement of size n_{gc} from a population of N_{gc} responses of which $N_{gc}p_{gc}$ are credited, so x_{gc} is hypergeometric. The argument has four steps.

Step one: sampling without replacement is dominated. Theorem 4 of Hoeffding (1963) states that if $X_1, \ldots, X_n$ is a sample drawn without replacement from a finite population and $Y_1, \ldots, Y_n$ is a sample of the same size drawn with replacement from the same population, then $\mathbb{E}\, f\big(\sum_i X_i\big) \le \mathbb{E}\, f\big(\sum_i Y_i\big)$ for every continuous convex f. Take $f(z) = e^{\lambda z}$, which is convex for every real λ. The with-replacement sum is binomial, so

$$\mathbb{E}\, e^{\lambda x_{gc}} \;\le\; \big(1 - p_{gc} + p_{gc} e^{\lambda}\big)^{n_{gc}}.$$

The inequality goes in the direction we need for both tails, because it holds for negative λ as well as positive.

Step two: the strata multiply. The G samples are drawn independently, so the moment generating function of x_c is the product of the individual ones: $\mathbb{E}\, e^{\lambda x_c} \le \prod_g (1 - p_{gc} + p_{gc} e^{\lambda})^{n_{gc}}$.

Step three: heterogeneous rates are dominated by their weighted mean. Let $\varphi(p) = \log\big(1 + p(e^{\lambda} - 1)\big)$, so that the product above is $\exp\big(\sum_g n_{gc}\, \varphi(p_{gc})\big)$. Writing $u = e^{\lambda} - 1$,

$$\varphi''(p) \;=\; \frac{-u^2}{(1+pu)^2} \;\le\; 0,$$

so φ is concave on $[0,1]$ for every λ. Jensen's inequality with the weights w_{gc}, which are nonnegative and sum to one, gives $\sum_g w_{gc}\,\varphi(p_{gc}) \le \varphi\big(\sum_g w_{gc}\,p_{gc}\big) = \varphi(\tilde{p}_c)$. Multiplying by n_c and exponentiating,

$$\mathbb{E}\,e^{\lambda x_c} \;\le\; \big(1-\tilde{p}_c+\tilde{p}_c e^{\lambda}\big)^{n_c}.$$

The heterogeneity has been absorbed: the bound is now the moment generating function of a single binomial at the weighted mean rate.

Step four: Chernoff. Fix $t>0$ with $\tilde{p}_c + t \le 1$. For $\lambda > 0$, Markov's inequality applied to $e^{\lambda x_c}$ gives

$$\Pr\big(\hat{p}_c \ge \tilde{p}_c + t\big) \;\le\; \exp\Big(-\lambda n_c(\tilde{p}_c+t) + n_c \log\big(1-\tilde{p}_c+\tilde{p}_c e^{\lambda}\big)\Big).$$

The exponent is minimised at $e^{\lambda^\star} = \frac{(\tilde{p}_c+t)(1-\tilde{p}_c)}{(1-\tilde{p}_c-t)\,\tilde{p}_c}$, which is larger than one and therefore admissible, and substituting it gives exponent $-n_c\,\mathrm{KL}(\tilde{p}_c + t \,\|\, \tilde{p}_c)$. This is the first inequality. For the second, apply the same four steps to the complementary counts $n_{gc} - x_{gc}$, whose rates are $1-p_{gc}$ and whose weighted mean is $1-\tilde{p}_c$, and use the identity $\mathrm{KL}(1-a\,\|\,1-b) = \mathrm{KL}(a\,\|\,b)$. □

The interval, and its coverage. Inverting the lemma gives the interval used in the paper. Define

$$[L_c, U_c] \;=\; \big\{q \in [0,1] \;:\; n_c\,\mathrm{KL}(\hat{p}_c \,\|\, q) \le \log(2/\alpha)\big\},$$

which is an interval containing $\hat{p}_c$, because $q \mapsto \mathrm{KL}(\hat{p}_c \,\|\, q)$ is convex with its minimum at $q = \hat{p}_c$.

Coverage of the interval. Let $t^\star$ solve $n_c\mathrm{KL}(\tilde{p}_c + t^\star \,\|\, \tilde{p}_c) = \log(2/\alpha)$. On the event $\{\tilde{p}_c < L_c\}$ we have $\hat{p}_c > \tilde{p}_c$ and $n_c\mathrm{KL}(\hat{p}_c \,\|\, \tilde{p}_c) > \log(2/\alpha)$. Since $a \mapsto \mathrm{KL}(a \,\|\, \tilde{p}_c)$ is increasing for $a > \tilde{p}_c$, this forces $\hat{p}_c > \tilde{p}_c + t^\star$, an event of probability at most $\alpha/2$ by Lemma 1. The event $\{\tilde{p}_c > U_c\}$ is bounded by $\alpha/2$ symmetrically, and the union bound gives coverage at least $1-\alpha$. □

Nothing in either argument requires the sub-population rates to be equal, requires sampling with replacement, or appeals to a limit. One sanity check is worth recording because it is exact. When no audited response in the cell is credited, $\hat{p}_c = 0$ and $\mathrm{KL}(0\,\|\,q) = -\log(1-q)$, so the upper endpoint solves $-n_c\log(1-U_c) = \log(2/\alpha)$, giving $U_c = 1-(\alpha/2)^{1/n_c}$. That is exactly the Clopper–Pearson upper limit at zero successes (Clopper & Pearson, 1934): at the boundary the design-valid interval costs nothing, and the price of validity is paid in the interior.

A.2 Proof of Proposition 1

The proof has three parts: the region contains the truth, the optimisation separates across cells, and the cellwise optimum has the stated closed form. A fourth part says where the region is sharp and where it is not.

The region contains the truth. Fix a concordant cell c. Model m's rate in that cell is the size-weighted mean of the rates of the sub-populations belonging to m, $p_{mc} = \sum_{g\in m}(N_{gc}/N_{mc})\,p_{gc}$, and $\tilde{p}_c$ is the design-weighted mean $\sum_g w_{gc}\,p_{gc}$ over all of them. Both are convex combinations of the same numbers $\{p_{gc}\}$. Under Assumption 1 those numbers lie in a common interval I of width at most ε, and a convex combination of points of I lies in I. Therefore p_{jc}, p_{kc} and $\tilde{p}_c$ all lie in I, which gives $|p_{jc}-p_{kc}| \le \varepsilon$ and $|p_{mc}-\tilde{p}_c| \le \varepsilon$ for $m \in \{j,k\}$. On the event $\tilde{p}_c \in [L_c, U_c]$ the second of these yields $p_{mc} \in [L_c-\varepsilon,\, U_c+\varepsilon]$, and since p_{mc} is a proportion it also lies in $[0,1]$, so $p_{mc} \in [\ell_c, h_c]$. Together these are exactly the constraints defining $R_c(\varepsilon)$. □

Separability. By Equation (1), $\Delta_{jk} = \sum_c (s_{jc}p_{jc} - s_{kc}p_{kc})$ is linear in the unknowns and the shares are known. The constraint set is a product across cells: each concordant cell contributes $R_c(\varepsilon)$, and each discordant cell contributes the rectangle $[l_{jc}, u_{jc}] \times [l_{kc}, u_{kc}]$, which is $[0,1]^2$ when that cell carries no labels for those models. Minimising or maximising a sum of terms over a product of sets is done term by term. For a discordant cell the optimum is at a corner, giving $s_{jc}l_{jc} - s_{kc}u_{kc}$ and $s_{jc}u_{jc} - s_{kc}l_{kc}$. □

The closed form. Write $a = s_{jc} \geq 0$, $b = s_{kc} \geq 0$, $\ell = \ell_c$, $h = h_c$, and minimise $f(x, y) = ax - by$ over $R = \{(x, y) \in [\ell, h]^2 : |x - y| \leq \varepsilon\}$.

For fixed $x \in [\ell, h]$, the objective decreases in y, so the optimal y is the largest feasible one, $y = \min(h,\, x + \varepsilon)$. This is feasible: it is at most h by construction, at least ℓ because $x \geq \ell$ and $h \geq \ell$, and within ε of x because it is at most $x + \varepsilon$ and at least $\min(h, x) = x$. Substituting,

$$g(x) \;=\; ax - b\min(h,\, x + \varepsilon) \;=\; \begin{cases} (a - b)\, x - b\, \varepsilon, & x \leq h - \varepsilon, \\ a\, x - b\, h, & x \geq h - \varepsilon, \end{cases}$$

a piecewise linear function whose slope rises from $a - b$ to a at the kink $x = h - \varepsilon$. Since $b \geq 0$ the slope is nondecreasing, so g is convex.

If $a \geq b$, both slopes are nonnegative and g is nondecreasing, so the minimum is at $x = \ell$ and equals $a\ell - b\min(h, \ell + \varepsilon)$.

If $a < b$, the first slope is negative and the second is nonnegative, so the unconstrained minimiser is the kink, and the constrained minimiser is $x^\star = \max(\ell,\, h - \varepsilon)$. When $h - \varepsilon \geq \ell$ the value is $a(h - \varepsilon) - bh$. When $h - \varepsilon < \ell$ the minimiser is ℓ, and since then $\ell + \varepsilon > h$ the value is $a\ell - bh$. Both cases are written by the single expression $a \max(\ell,\, h - \varepsilon) - b\, h$.

The maximum follows by symmetry. R is invariant under exchanging its two coordinates, and $\max_R(ax - by) = -\min_R(bx - ay)$ after that exchange, so the maximum is obtained by applying the formula above with a and b swapped and negating the result. □

Sharp at $\varepsilon = 0$, conservative above it. At $\varepsilon = 0$ we have $\ell_c = L_c$ and $h_c = U_c$, and $R_c(0)$ is the diagonal segment $\{(q, q) : q \in [L_c, U_c]\}$. Every point of it is attained in the relaxation where each p_{gc} may take any value in $[0, 1]$: set every sub-population rate in the cell to q, which satisfies $H(0)$ and makes $p_{jc} = p_{kc} = \tilde{p}_c = q$. The interval for Δ_{jk} is therefore attained and cannot be narrowed, which is what sharpness means here. The qualification about the relaxation matters, because each p_{gc} is really a fraction over finitely many responses and only finitely many values of q are exactly attainable. This is the same granularity that Section 2 uses to argue that $H(0)$ is an idealisation rather than a hypothesis about the world.

For $\varepsilon > 0$ the region is an outer bound, and the reason is that it forgets the weights. $R_c(\varepsilon)$ imposes only that p_{jc} and p_{kc} lie within ε of each other inside $[\ell_c, h_c]$. The actual constraint is stronger: the same sub-population rates that produce p_{jc} and p_{kc} must also produce a design-weighted mean $\tilde{p}_c$ lying in $[L_c, U_c]$. When one model supplies most of the audit weight in the cell, its own rate is pinned close to $\tilde{p}_c$ and cannot travel the full distance ε away from $[L_c, U_c]$. The computed range therefore contains the sharp identified range. Every consequence runs in the safe direction: reported ranges are at least as wide as the truth requires, fewer orderings are certified than a sharp analysis would certify, and each breakdown value in Definition 1 is a lower bound on the sharp one. □

A.3 Proof of Proposition 2

Proof of Proposition 2. Let a_m be the fraction of model m's responses the single verifier credits. Conditioning on the verifier's verdict and using the assumed common predictive values,

$$A_m \;=\; \Pr(Y^* = 1 \mid v = 1)\, a_m + \Pr(Y^* = 1 \mid v = 0)\, (1 - a_m) \;=\; \pi_0 + a_m(\pi_1 - \pi_0),$$

so $\Delta_{jk} = A_j - A_k = (\pi_1 - \pi_0)(a_j - a_k)$. □

The same statement in the other natural parameterisation makes the consequence easier to see. Suppose instead that the verifier has common error rates, $q = \Pr(v = 1 \mid Y^* = 0)$ and $r = \Pr(v = 0 \mid Y^* = 1)$, shared by all models. Then $a_m = (1 - r)A_m + q(1 - A_m) = q + (1 - q - r)A_m$, so $a_j - a_k = (1 - q - r)\, \Delta_{jk}$. The observed gap is the human gap scaled by one number.

Either way, a single multiplicative constant relates every observed pairwise gap to the corresponding human one. Three consequences follow, and they are the reason the design uses more than one verifier.

The sign of every pairwise comparison is determined by the sign of that constant. When $\pi_1 > \pi_0$, equivalently $q + r < 1$, every ordering the verifier reports is the human ordering. When $\pi_1 < \pi_0$ the whole board is reversed at once. A model that a common mechanism treats unusually badly does not exist inside this parameterisation.

Pair-specific reversals are therefore outside the model. A reversal that affects one pair and leaves another intact cannot be produced by any choice of π_0 and π_1, so observing one is evidence that the mechanism differs across models. Section 5 reports such reversals.

At $\pi_1 = \pi_0$, equivalently $q + r = 1$, every Δ_{jk} is zero and the verifier carries no information about the ordering at all.

The role this plays in the design is the following. Pooling under a single verifier is Assumption 1 at $\varepsilon = 0$ with $K = 1$, where the pooled cells cover the entire population and nothing is left model-specific. That assumption does not merely simplify the analysis; by the proposition it removes the phenomenon the paper is about. With K verifiers the pooling is confined to the cells where they agree, and the cells where they disagree, which is where the disputed responses live, stay model-specific by construction.

A.4 Proof of Proposition 3

The audit half, in finite samples. List the parameters a certification uses: the pooled rate $\tilde{p}_c$ for each concordant cell, which is shared by every pair on the board, and the per-model rate p_{mc} for each audited discordant cell. Distinct parameters are estimated from disjoint sets of labels, drawn independently by the design, so the corresponding intervals are functions of independent random variables. Give each of the P parameters an interval at level $1 - \alpha'$ with $\alpha' = 1 - (1 - \alpha_{\text{audit}})^{1/P}$, which is the allocation of Šidák (1967) and is exact rather than conservative under independence. By Lemma 1 each interval covers its parameter with probability at least $1 - \alpha'$, and by independence all of them cover simultaneously with probability at least $(1 - \alpha')^P = 1 - \alpha_{\text{audit}}$.

On that event, the hypothesis of Proposition 1 holds for every cell at once, so the identified range computed for every pair contains the corresponding Δ_{jk} simultaneously. Any ordering whose range lies strictly above zero is therefore correct. No step used a limit, so the statement is exact in finite samples. □

The item-sampling half, asymptotically. Let δ_{jk} denote the pairwise difference over the distribution from which the benchmark's items were drawn, and Δ_{jk} the difference on the realised items. With D_i the difference of the two models' human-reference verdicts on item i, the sample mean of D is Δ_{jk} and its expectation is δ_{jk}. The central limit theorem gives $\Pr(|\delta_{jk} - \Delta_{jk}| > z_{1-\alpha_s/2}\sqrt{\text{Var}(D)/N}) \to \alpha_s$, and replacing $\text{Var}(D)$ by the upper bound of Section 3 can only widen the interval. Combining the two halves by a union bound,

$$\Pr\big(\delta_{jk} \notin [\text{LB}_{jk} - h_{jk},\, \text{UB}_{jk} + h_{jk}]\big) \;\le\; \Pr\big(\Delta_{jk} \notin [\text{LB}_{jk}, \text{UB}_{jk}]\big) + \Pr\big(|\delta_{jk} - \Delta_{jk}| > h_{jk}\big),$$

which is at most $\alpha_{\text{audit}} + \alpha_{\text{sample}}$. The two events concern different sources of error, one the labels and the other the items, so nothing is counted twice; adding the levels rather than convolving the two error distributions is what makes the combination conservative. □

For a family of orderings, the audit half is already simultaneous because the concordant intervals are shared and the discordant allocation runs over all audited per-model parameters. The sampling half is made simultaneous by a Bonferroni division of α_{sample} across the family. Rank intervals inherit the family level directly: on the event that every certified ordering is correct, at least one model is ranked above m for each ordering $k \succ m$ that was certified and at least one below for each $m \succ k$, so m's true rank lies between one plus the first count and the number of models less the second.

A.5 The paired variance bound

Section 3 bounds the item-sampling variance without observing the human reference on unaudited items. Two elementary steps do the work.

First, D_i takes values in $\{-1, 0, 1\}$, so D_i^2 is the indicator that the two models are scored differently and $\mathrm{Var}(D) \le \mathbb{E}[D^2] = \Pr(D \neq 0)$.

Second, for any verifier v and any response pair,

$$\mathbf{1}[Y_j^* \neq Y_k^*] \;\le\; \mathbf{1}[v_j \neq v_k] + \mathbf{1}[Y_j^* \neq v_j] + \mathbf{1}[Y_k^* \neq v_k].$$

If the left side is one and all three terms on the right are zero, then $v_j = v_k$, $Y_j^* = v_j$ and $Y_k^* = v_k$, which forces $Y_j^* = Y_k^*$ and contradicts the left side. Averaging over items, the disagreement rate between the two models under the human reference is at most the verifier's own disagreement rate between them, which is observed on every item, plus each model's disagreement rate with that verifier, which the identified bounds of Proposition 1 bound above at the same ε. Taking the smallest bound over the available verifiers, and capping at one, gives the quantity used in the half-width. It is paired, because the verifier disagreement term is computed item by item rather than from the two models' marginal accuracies, and it is conservative, because every ingredient is an upper bound.

B Numerical checks of the stated results

Appendix A proves the results. This appendix describes what is checked numerically, which is a different question with three parts: whether the code computes what the propositions say, whether the prose says what the code computes, and whether a change to the data pipeline silently moves a number that was published earlier. The released suite holds 880 cases in 71 functions, counted by a script so that this sentence does not have to be maintained by hand.

B.1 The propositions against independent computation

`tests/test_certification.py` contributes 613 cases. Each compares the implementation against a computation that does not share its code path.

The closed form of Proposition 1 is checked against brute force on randomly drawn cells: a dense grid over $R_c(\varepsilon)$, together with the region's exact polygon vertices so that an optimum sitting between grid points cannot be missed. A separate check asserts where the minimiser sits, not only what the minimum equals, because a formula can return the right value from the wrong argument and stay wrong at the next input. Three degenerate configurations are pinned explicitly: the cell where no verifier credits, in which $L_c = 0$ and the clipping binds for every positive ε; a cell in which both models have zero share and must contribute nothing; and an ε large enough that the region is the whole unit square.

Monotonicity of the lower bound in ε is checked over a grid, since Definition 1 is well posed only if it holds. The breakdown value itself is checked by running the bisection and, separately, scanning a fine grid for the first ε at which the margin fails; the two must agree to a stated tolerance. The interior slope of a concordant cell's contribution is checked to be $-\max(s_{jc}, s_{kc})$. That check exists because an earlier version of the formula used $-(s_{jc} + s_{kc})$, which is valid but looser, and nothing in the output would have said so.

Proposition 2 is checked in both parameterisations over random draws, for every pair in a synthetic board, which is the form in which the claim is actually used: not that some pair is scaled, but that all of them are scaled by the same number. The falsification power calculation is checked by inverting it, and the claim that sampling without replacement can only help is checked against exact binomial coefficients rather than against the bound that replaces them.

Lemma 1 gets three checks. At zero and at full successes the interval must equal the Clopper–Pearson limit exactly, which is the boundary identity derived in Appendix A. The interval must contain the point estimate and must widen with the level. And a Monte Carlo draws pooled hypergeometric samples from sub-populations with deliberately different rates and requires the realised miss rate not to exceed nominal. That last one is the only check that exercises both departures the lemma is about at once, and it is the check that would fail if the Jensen step were wrong.

Rank intervals are checked against a small board whose certified set and correct answer are written out by hand.

B.2 The prose against the code

`tests/test_manuscript_claims.py` contributes 203 cases and exists because transcription is where errors enter a paper. The formula as printed in Section 2 is re-implemented from the LaTeX text, deliberately without importing the library, and compared against the implementation on random inputs. The same is done for Proposition 2, for the claim that $\varepsilon = 0$ collapses to the pooled analysis, and for the claim that certification is monotone. If an edit to the manuscript changes a formula and the implementation is not changed with it, the suite fails rather than the reviewer noticing.

B.3 The regression anchor

`tests/test_regression_anchor.py` contributes 7 cases. It rebuilds the earlier seventeen-model, two-verifier configuration from the *current* census and audit map and requires it to reproduce the counts published from the earlier pipeline: 84 orderings combined and 101 audit-only, under the historical string semantics. The point is not the old numbers. It is that any change to the census, the audit map or the verifiers that silently alters a previously published result fails the build. The defect described in Section 4, in which a verifier degraded to string comparison without saying so, is exactly the class of failure this check is for.

B.4 What these checks do not establish

They do not prove the propositions; Appendix A does. They do not validate the empirical results, which come from the committed result files and the figures generated from them. And a check written from the same misunderstanding as the code will pass. That is the reason for the division of labour above: the propositions are proved on paper, the manuscript-claims layer re-implements from the printed text rather than from the library, and the anchor compares against numbers fixed before the current code existed.

C Inference details

C.1 How the levels are allocated

The audit and the item sample are controlled separately, at $\alpha_{\text{audit}} = 0.025$ and $\alpha_{\text{sample}} = 0.025$, and a claim that generalises beyond the benchmark's items is made at one minus their sum. Proposition 3 is what licenses adding them.

Within the audit, the level is divided across the parameters a claim uses, by the allocation $\alpha' = 1 - (1 - \alpha_{\text{audit}})^{1/P}$. For a single ordering those parameters are the pooled interval of each concordant cell, plus each of the two models' audited discordant cells. In the configuration reported here no discordant cell carries labels, so P is the number of concordant cells and every pooled interval is constructed at level 0.9874.

Making a statement about the whole board changes less than it might appear. The pooled concordant intervals are shared by every pair, so the audit side of a family-wide statement is already simultaneous in the present configuration and needs no further widening. What the family-wide setting changes is the sampling term, which takes a Bonferroni division of α_{sample} across the 351 orderings. The specification for the general case is worth stating because it is not the same: once discordant cells carry labels, a per-model interval enters every ordering that model appears in, and the allocation must then run over all audited per-model parameters on the board rather than over the parameters of one pair.

Rank intervals are read off the certified set and inherit whichever level that set was constructed at, since they are a deterministic function of it.

C.2 A sampling term that does not rest on a limit

The paired half-width of Section 3 is conservative in its variance but asymptotic in its distribution, and a reader who declines the central limit approximation should be given a number rather than a caveat. Because the paired difference D_i lies in $[-1, 1]$, Hoeffding's inequality bounds the deviation of its mean without any

variance estimate at all, giving the half-width

$$h \;=\; \sqrt{\frac{2\log(2/\alpha_{\text{sample}})}{N}},$$

which is finite-sample, distribution-free, and the same for every pair. It is also much wider, and the cost is visible: substituting it certifies 100 orderings where the paired asymptotic term certifies 153. Both are reported. The distribution-free figure is the one that survives a reader who rejects the approximation, and the gap between them is the price of that approximation, stated rather than assumed away.

The pairing matters separately from the distributional question. An earlier unpaired term, built from each model's marginal accuracy under one verifier, is retained in the code only to reproduce the regression anchor of Appendix B; it certifies 160 orderings. It is not used, because two models answering the same items differ on far fewer of them than their marginal accuracies suggest, and a term that ignores the pairing throws that away.

C.3 What the coverage simulation does

The simulation in `scripts/fb_certify_coverage_sim.py` runs 9 scenarios at 2,000 replicates each, at the full scale of the board. It checks whether the guarantee is attained and how it degrades; it is not a proof, and Section 3 says so.

The frame is the real census, with the real cell structure and the real audit allocation, so that the strata, their sizes and their sampling rates are the ones the paper actually uses. What must be invented is the human reference on unaudited responses, and it is invented in two deliberately different ways.

In the first, each concordant sub-population is given a number of errors that makes the realised error fractions span an interval of width exactly $\varepsilon_{\text{true}}$, so that Assumption 1 holds exactly as a statement about finite populations. This is the validity check: the procedure is analysed at the same ε, and no certified claim should be wrong.

In the second, concordant errors are drawn independently with a small propensity that is the same for every model. No model differs from any other in expectation, which is the most favourable situation pooling could hope for, and yet the realised fractions differ by chance, so $H(0)$ fails as a statement about the finite populations that actually exist. This is a diagnostic rather than a validity check, and it is the one that produced the paper's reason for not reporting $\varepsilon = 0$ as a result.

Discordant cells are given independent per-model rates drawn uniformly, which is the worst case the analysis is built for and the reason those cells enter at $[0, 1]$ when unaudited. Items are resampled from the frame with replacement and paired across models, so the item-sampling term has something real to cover. An audit-only claim is scored against the difference on the realised items; a combined claim is scored against the superpopulation expectation. Confusing those two would make the procedure look better than it is, which is why they are kept apart in the output.

One number bounds what the simulation can resolve. At the nominal level and this many replicates, the Monte Carlo standard error of a family-wise error rate is 0.003491, so a reported rate of zero means below resolution rather than exactly zero, and the scenarios are compared against nominal with that in mind.

D Sensitivity analyses

[TO WRITE: the registered sensitivity table from `certify_results.json`.]

E Sensitivity to the human reference routine

[TO WRITE: string, harness and Math-Verify references compared.]